Conjecture 00000003718: the matching polynomial of a tree has only real roots

A proof, with a Lean 4 / Mathlib formalization

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27 October 2026

Abstract

We prove conjecture 00000003718 of *The Last Math Competition*: for every finite tree T the matching polynomial $\mu(T;x)=\sum_{k\ge 0}(-1)^k m_k(T)\,x^{n-2k}$ has only real roots, and the distribution of these roots is determined by the walk counts of T. The proof goes through the identity $\mu(F;x)=\det(xI-A(F))$ for every forest F, established by a leaf-deletion induction on both sides, followed by the spectral theorem for real symmetric matrices; the moment identity $\sum_i \theta_i^k = \operatorname{tr} A(T)^k = \#\{\text{closed walks of length } k\}$ identifies the root distribution with the moment transform of the closed-walk counts. Every statement is formalized and machine-checked in Lean 4 (toolchain `v4.33.1`, Mathlib `v4.33.1`); the project compiles with no `sorry`, no `native_decide` and no axioms beyond `propext`, `Classical.choice` and `Quot.sound`.

1 The conjecture as filed

English. Definition: Matching polynomials of trees. Conjecture: The real roots of matching polynomials: the roots of the matching polynomial of trees are all real, with the distribution of real roots the transform of path counts. (tree matching real root path count transform distribution)

(The Chinese version of the statement in `conjectures/00000003718.md` is a literal translation of the English one.)

1.1 Reading of the statement

Throughout, a *graph* is a finite simple graph $G=(V,E)$ with $n=|V|$ vertices; a *tree* is a connected acyclic graph, a *forest* an acyclic graph.

Definition 1.1 (matching polynomial)**.** A *k-matching* of G is a set of k pairwise vertex-disjoint edges; write $m_k(G)$ for the number of k-matchings ($m_0=1$). The *matching polynomial* of G is

$$\mu(G;x) \;=\; \sum_{k\ge 0}(-1)^k\, m_k(G)\, x^{\,n-2k} \;=\; \sum_{M \text{ matching of } G} (-1)^{|M|} x^{\,n-2|M|} \in \mathbb{Z}[x].$$

This is the standard definition (Heilmann–Lieb 1972; Farrell 1979; Godsil–Gutman 1981), and it is exactly the one formalized in `SimpleGraph.matchingPolynomial`; the equality of the two displayed forms is the Lean theorem `matchingPolynomial_eq_sum_numMatchings`.

The conjecture has two clauses.

(C1) *"the roots of the matching polynomial of trees are all real"*: for every tree T, every (complex) root of $\mu(T;x)$ is real; equivalently $\mu(T;x)$ splits into n linear factors over $\mathbb{R}$.

(C2) *"with the distribution of real roots the transform of path counts"*: the multiset of roots $\{\theta_1, \dots, \theta_n\}$ is determined by counting walks in T. We make this precise through its moments: for every $k \geq 0$,

$$\sum_{i=1}^{n} \theta_i^k \;=\; \#\{\text{closed walks of length } k \text{ in } T\} \;=\; \sum_{v \in V} \#\{\text{walks } v \to v \text{ of length } k\}.$$

The power sums of a degree-n monic polynomial determine it (Newton's identities), so the root distribution is literally the moment transform of the closed-walk counting sequence; in a tree every closed walk is a back-and-forth traversal of a path, which is the "path count" phrasing of the statement.

We prove both clauses. The precise formal statement is Theorem 2.2 below, whose Lean counterpart is `TLMC3718.conjecture_00000003718`.

Remark 1.2. Clause (C1) is a classical theorem: Heilmann and Lieb (1972) proved real-rootedness of $\mu(G; x)$ for *every* graph, and Godsil and Gutman (1981) observed that for forests $\mu(F; x) = \det(xI - A(F))$, which gives the tree case directly. We give a complete self-contained proof of the tree case along the second route, and we formalize it; we are not aware of a previous formalization of these results in Mathlib.

2 Statement of the results

Let $A = A(G) \in \{0,1\}^{V \times V}$ be the adjacency matrix and $\mathrm{charpoly}(A) = \det(xI - A) \in \mathbb{Z}[x]$. For $S \subseteq V$ let $G[S]$ be the induced subgraph, A_S its adjacency matrix (the principal submatrix of A indexed by S) and $\mu_S(x) = \mu(G[S]; x)$; a matching of $G[S]$ is a set of edges of G with both end points in S, pairwise disjoint.

Theorem 2.1 (forests: $\mu = \mathrm{charpoly}$)**.** *Let G be a forest. For every $S \subseteq V$, $\mu(G[S]; x) = \det(xI - A_S)$. In particular $\mu(G; x) = \mathrm{charpoly}(A(G))$.*

Theorem 2.2 (main theorem; conjecture 00000003718)**.** *Let T be a tree on n vertices with roots $\theta_1, \dots, \theta_n$ of $\mu(T; x)$ in $\mathbb{C}$ (with multiplicity). Then*

(a) $\mu(T; x)$ splits over $\mathbb{R}$: $\mu(T; x) = \prod_{i=1}^{n}(x - \theta_i)$ with all $\theta_i \in \mathbb{R}$; in particular $\mu(T; x)$ has exactly n real roots counted with multiplicity, and every complex root z of $\mu(T; x)$ satisfies $\mathrm{Im}\, z = 0$;

(b) for every $k \in \mathbb{N}$, $\sum_{i=1}^{n} \theta_i^k = \sum_{v \in V} \#\{\text{walks } v \to v \text{ in } T \text{ of length } k\}$.

3 Proofs

3.1 Leaves of forests

Lemma 3.1. *Let G be a forest and $S \subseteq V$. If G has an edge with both end points in S, then there is $v \in S$ having exactly one neighbour in S.*

Proof. $G[S]$ is again a forest, and it has an edge, so it has a path of length ≥ 1. Let P be a path of maximal length in $G[S]$, with first vertex v and second vertex u. If $w \in S$ is a neighbour of v with $w \notin P$, then prepending w to P gives a longer path, contradicting maximality; so every neighbour of v in S lies on P. In an acyclic graph the only vertex of P adjacent to its first vertex v is the second vertex u (otherwise the subpath of P from v to w together with the edge wv is a cycle). Hence u is the unique neighbour of v in S. □

3.2 The leaf recurrence for the matching polynomial

Lemma 3.2. *Let $S \subseteq V$, $v \in S$, and suppose $u \in S$ is the unique neighbour of v in S. Then*

$$\mu_S(x) \;=\; x\,\mu_{S\setminus\{v\}}(x) \;-\; \mu_{S\setminus\{v,u\}}(x).$$

Proof. Split the matchings M of $G[S]$ according to whether $vu \in M$. If $vu \notin M$ then no edge of M contains v (such an edge would be vw with $w \in S$ a neighbour of v, hence $w = u$), so M is a matching of $G[S \setminus \{v\}]$; conversely every matching of $G[S \setminus \{v\}]$ is a matching of $G[S]$ not containing vu. Since $|S| = |S \setminus \{v\}| + 1$ and $2|M| \le |S \setminus \{v\}|$ (a matching inside a set S' covers $2|M|$ distinct vertices of S'), these matchings contribute $\sum_M (-1)^{|M|} x^{|S\setminus\{v\}|+1-2|M|} = x\,\mu_{S\setminus\{v\}}(x)$. If $vu \in M$ then $M' = M \setminus \{vu\}$ is a matching of $G[S \setminus \{v,u\}]$ (its edges are disjoint from vu), and $M \mapsto M'$ is a bijection onto the matchings of $G[S \setminus \{v,u\}]$ with inverse $M' \mapsto M' \cup \{vu\}$; as $|S| = |S \setminus \{v,u\}| + 2$ and $|M| = |M'| + 1$, these contribute $\sum_{M'} (-1)^{|M'|+1} x^{|S\setminus\{v,u\}|-2|M'|} = -\mu_{S\setminus\{v,u\}}(x)$. □

3.3 The leaf expansion of a determinant

Lemma 3.3. *Let R be a commutative ring, I a finite index set, $N \in R^{I\times I}$, and $v \neq u$ in I such that $N_{vw} = 0$ and $N_{wv} = 0$ for all $w \notin \{v,u\}$. Writing $N^{(v)}$ and $N^{(v,u)}$ for the principal submatrices obtained by deleting v, resp. v and u,*

$$\det N \;=\; N_{vv}\,\det N^{(v)} \;-\; N_{vu}N_{uv}\,\det N^{(v,u)}.$$

Proof. Row v of N equals $N_{vv}e_v + N_{vu}e_u$, so by linearity of the determinant in row v, $\det N = N_{vv}\det N_1 + N_{vu}\det N_2$, where N_1 (resp. N_2) is N with row v replaced by e_v (resp. e_u). *(i)* In N_1 the row v is e_v; after moving v to the first position, N_1 is block lower triangular with blocks (1) and $N^{(v)}$, so $\det N_1 = \det N^{(v)}$. *(ii)* In N_2 the column v equals $N_{uv}e_u$ (its entry in row v is $(e_u)_v = 0$, and the other entries of column v are those of N). By linearity in column v, $\det N_2 = N_{uv}\det P$, where P is N_2 with column v replaced by e_u; so row v and column v of P both equal e_u. Swapping rows u and v of P gives a matrix Q with $\det Q = -\det P$ whose row u is e_u; expanding as in (i), $\det Q = \det Q^{(u)}$. In $Q^{(u)}$ the column v is e_v (its entry in row v is $P_{uv} = 1$, the others are $P_{wv} = 0$ for $w \neq u, v$), so expanding along column v, $\det Q^{(u)} = \det Q^{(u,v)}$, and the remaining matrix $Q^{(u,v)}$ is exactly $N^{(v,u)}$ (its entries avoid rows and columns u, v, where Q agrees with N). Altogether $\det N_2 = N_{uv}\cdot(-\det N^{(v,u)})$, which gives the claim. □

Lemma 3.4. *With S, v, u as in Lemma 3.2,* $\det(xI - A_S) = x\det(xI - A_{S\setminus\{v\}}) - \det(xI - A_{S\setminus\{v,u\}})$.

Proof. Apply Lemma 3.3 to $N = xI - A_S$ over $R = \mathbb{Z}[x]$ (or any commutative ring). Row v of A_S has a single 1, in column u, and A_S is symmetric, so the support hypotheses hold; $N_{vv} = x$ (there are no loops), $N_{vu} = N_{uv} = -1$, and $(xI - A_S)^{(v)} = xI - A_{S\setminus\{v\}}$, $(xI - A_S)^{(v,u)} = xI - A_{S\setminus\{v,u\}}$. □

3.4 Proof of Theorem 2.1

By strong induction on $|S|$. If G has no edge inside S then the only matching of $G[S]$ is $\emptyset$, so $\mu_S(x) = x^{|S|}$, while $A_S = 0$ and $\det(xI - 0) = x^{|S|}$. Otherwise Lemma 3.1 provides $v \in S$ with a unique neighbour $u \in S$, and Lemmas 3.2 and 3.4 give the same recurrence for μ_S and for $\det(xI - A_S)$ in terms of the strictly smaller sets $S \setminus \{v\}$ and $S \setminus \{v,u\}$, where the two agree by induction. The case $S = V$ is the statement for G. □

3.5 Proof of Theorem 2.2

A tree is a forest, so $\mu(T;x) = \text{charpoly}(A)$ with $A = A(T)$ real symmetric (Theorem 2.1, over $R = \mathbb{R}$ and over $R = \mathbb{C}$; the matching polynomial is compatible with change of coefficient ring). By the spectral theorem $A = U\,\mathrm{diag}(\lambda_1, \ldots, \lambda_n)\,U^*$ with U orthogonal and $\lambda_i \in \mathbb{R}$, hence $\text{charpoly}(A) = \prod_i (x - \lambda_i)$ splits over $\mathbb{R}$, has exactly $n = \deg \text{charpoly}(A)$ roots with multiplicity, and its roots in $\mathbb{C}$ are the real numbers λ_i: this is (a). For (b), $\sum_i \lambda_i^k = \operatorname{tr} \mathrm{diag}(\lambda_i)^k = \operatorname{tr}(U \mathrm{diag}(\lambda_i)^k U^*) = \operatorname{tr} A^k = \sum_v (A^k)_{vv}$, and $(A^k)_{vw}$ is the number of walks of length k from v to w. □

4 The Lean 4 formalization

The directory `lean/` is a Lake project (toolchain `leanprover/lean4:v4.33.1`, Mathlib tag `v4.33.1`, revision `0df444a360eaa60ab8c11dca51a86af692955474`).

File	Content
`Submission/DetLeaf.lean`	Lemma 3.3: `TLMC3718.det_leaf`
`Submission/MatchingPoly.lean`	definitions; Lemma 3.2: `matchingPolyIn_leaf`
`Submission/Leaf.lean`	Lemma 3.1: `IsAcyclic.exists_leaf_in`
`Submission/Charpoly.lean`	Lemma 3.4; Theorem 2.1: `matchingPolynomial_eq_charpoly`
`Submission/Main.lean`	Theorem 2.2: `TLMC3718.conjecture_00000003718`
`Submission/Verification.lean`	`#print axioms` of every result

Definitions. For `G : SimpleGraph V` with `[Fintype V] [DecidableEq V] [DecidableRel G.Adj]` and `s : Finset V` (all in namespace `SimpleGraph`):

```
def IsMatchingIn (s : Finset V) (M : Finset (Sym2 V)) : Prop :=
  M ⊆ G.edgeFinset ∧ (∀ e ∈ M, ∀ x, x ∈ e → x ∈ s) ∧
    ∀ e ∈ M, ∀ f ∈ M, e ≠ f → ∀ x, x ∈ e → x ∉ f
noncomputable def matchings (s : Finset V) : Finset (Finset (Sym2 V)) :=
  G.edgeFinset.powerset.filter (G.IsMatchingIn s)
noncomputable def numMatchings (s : Finset V) (k : ℕ) : ℕ :=
  ((G.matchings s).filter (fun M => M.card = k)).card
noncomputable def matchingPolyIn (s : Finset V) : R[X] :=
  ∑ M ∈ G.matchings s, (-1 : R[X]) ^ M.card * X ^ (s.card - 2 * M.card)
noncomputable def matchingPolynomial : R[X] := G.matchingPolyIn R Finset.univ
def adjMatrixIn (s : Finset V) : Matrix s s R :=
  (G.adjMatrix R).submatrix Subtype.val Subtype.val
```

So `G.matchingPolynomial R` is $\mu(G;x)$ with coefficients in the commutative ring R; the theorem
`matchingPolynomial_eq_sum_numMatchings` proves that it equals $\sum_{k=0}^{n} (-1)^k\, m_k\, X^{n-2k}$ with m_k =`numMatchings univ k`, the classical form. Trees are Mathlib's `SimpleGraph.IsTree` (connected and acyclic); walks are Mathlib's `SimpleGraph.Walk`.

```
theorem SimpleGraph.matchingPolynomial_eq_charpoly (hG : G.IsAcyclic) :
    G.matchingPolynomial R = (G.adjMatrix R).charpoly

theorem TLMC3718.conjecture_00000003718 (T : SimpleGraph V) [DecidableRel T.Adj]
    (hT : T.IsTree) :
    (T.matchingPolynomial ℝ).Splits ∧
    (T.matchingPolynomial ℝ).roots.card = Fintype.card V ∧
    (∀ z : ℂ, (T.matchingPolynomial ℂ).IsRoot z → z.im = 0) ∧
    (∀ k : ℕ, ((T.matchingPolynomial ℝ).roots.map (fun r => r ^ k)).sum =
      ∑ v, (Fintype.card {p : T.Walk v v | p.length = k} : ℝ))
```

Here `Polynomial.Splits` is Mathlib's predicate "product of a constant and linear factors"; the four conjuncts are Theorem 2.2(a) (three forms) and (b). The individual results are `matchingPolynomial_splits`, `card_roots_matchingPolynomial`, `exists_real_of_isRoot` / `im_eq_zero_of_isRoot` and `sum_roots_pow_eq_card_closedWalks`.

Audit. `lake build` succeeds; the sources contain no `sorry`, `native_decide`, `axiom`, `unsafe`, `opaque`, `implemented_by` or `extern`. `Submission/Verification.lean` prints, for every theorem listed above,

```
'TLMC3718.conjecture_00000003718' depends on axioms: [propext, Classical.choice, Quot
    .sound]
```

Reproduction. In `lean/`: `lake exe cache get && lake build`. This report compiles with `tectonic report.tex` (or `pdflatex`).

References

- O. J. Heilmann, E. H. Lieb, *Theory of monomer–dimer systems*, Comm. Math. Phys. 25 (1972), 190–232.
- C. D. Godsil, I. Gutman, *On the theory of the matching polynomial*, J. Graph Theory 5 (1981), 137–144.
- C. D. Godsil, *Algebraic Combinatorics*, Chapman & Hall, 1993, Ch. 1 and 6.